AI & MACHINE LEARNING CHALLENGE

Industrial Chemical Reactor Yield Prediction

A Machine Learning Surrogate Framework for Continuous Non-Isothermal Reactors

TEAM NAME

Saubhagya se Prem

FOCUS AREA

Process Optimization & Surrogate Modeling

Problem Statement

THE INDUSTRIAL CHALLENGE

Physics Simulation Bottlenecks

Modern chemical manufacturing relies on complex continuous non-isothermal reactor simulations governing reaction kinetics, fluid dynamics, and heat transfer.

Desired Reaction: **A → B** | **Side Reaction:** **B → C**

Although exact physics-based differential solvers yield high fidelity, they are computationally expensive, slow, and unsuitable for real-time control systems.

OPERATIONAL IMPACT

The Operational Need

Plant operators and automated SCADA systems require instantaneous feedback to continuously adjust operating parameters and minimize side-product formation.

Delay in yield predictions leads to sub-optimal operating points, increased energy consumption, raw material waste, and off-spec product loss.

Goal: Replace physics solvers with an instantaneous ML surrogate model predicting **Overall Yield of Product B**.

Proposed Solution

Data-Driven Surrogate Framework

We propose a fast, accurate regression-based machine learning framework that learns the complex nonlinear mapping directly from reactor operating variables to target yield.

- **Instantaneous Predictions:** Millisecond latency enabling real-time closed-loop decision support.
- **Non-Isothermal Mechanics:** Captures thermal jacket coupling and exothermic heat generation.
- **Robust Generalization:** Verified across multi-fold cross-validation with synthetic noise tolerance.

Domain-Inspired Feature Engineering

Thermal Metrics

Temperature Difference: Measures thermal gradients between jacket and inlet fluid.

$$\Delta T = T_{jacket} - T_{inlet}$$

Drives heat transfer kinetics.

Residence Time Index

Residence Time Proxy: Ratio of reactor bed length to fluid velocity.

$$\tau = \frac{length\_m}{flow\_rate}$$

Controls contact time & selectivity.

Thermal Energy Index

Energy Interaction: Non-linear interaction between thermal states.

$$E_{idx} = T_{inlet} \times T_{jacket}$$

Key predictor of activation energy.

Model Benchmarks (RMSE)

Benchmark Insight: Gradient Boosting achieved the lowest RMSE (13.687) and highest variance explanation (89.3%), closely followed by tuned XGBoost. Combining both via weighted ensemble yields optimal stability.

Uniqueness & Approach Feasibility

89.3%

Yield Variance Explained (R^2)

What Makes Our Approach Unique?

Unlike standard off-the-shelf ML models, our pipeline bridges physical chemical kinetics with machine learning through three distinct innovations:

- **Physics-Aware Features:** Engineered residence time and thermal energy interaction terms that reflect Arrhenius behavior.
- **Ensemble Averaging:** Combines top gradient boosted decision tree architectures to prevent overfitting.
- **SHAP Transparency:** Complete black-box breakdown ensuring chemical process safety and trust.

SHAP Feature Explainability

- **Thermal Energy Index Dominance (~46% Importance):** SHAP feature importance analysis confirms that the engineered thermal energy interaction term ($T_{inlet} \times T_{jacket}$) is the single strongest driver of final yield prediction.
- **Inlet Temperature Impact (~23% Importance):** High inlet temperatures exponentially accelerate activation energy for reaction $A \rightarrow B$, but require precise jacket temperature control to suppress the side reaction ($B \rightarrow C$).
- **Residence Time Tuning (~10% Importance):** Optimal contact time ensures complete conversion of reactant A without over-exposing product B to unwanted side-decomposition.
- **Operator Transparency:** SHAP force plots provide operators with actionable feedback on which physical parameter needs adjustment during drifting runs.

Technology Stack Architecture

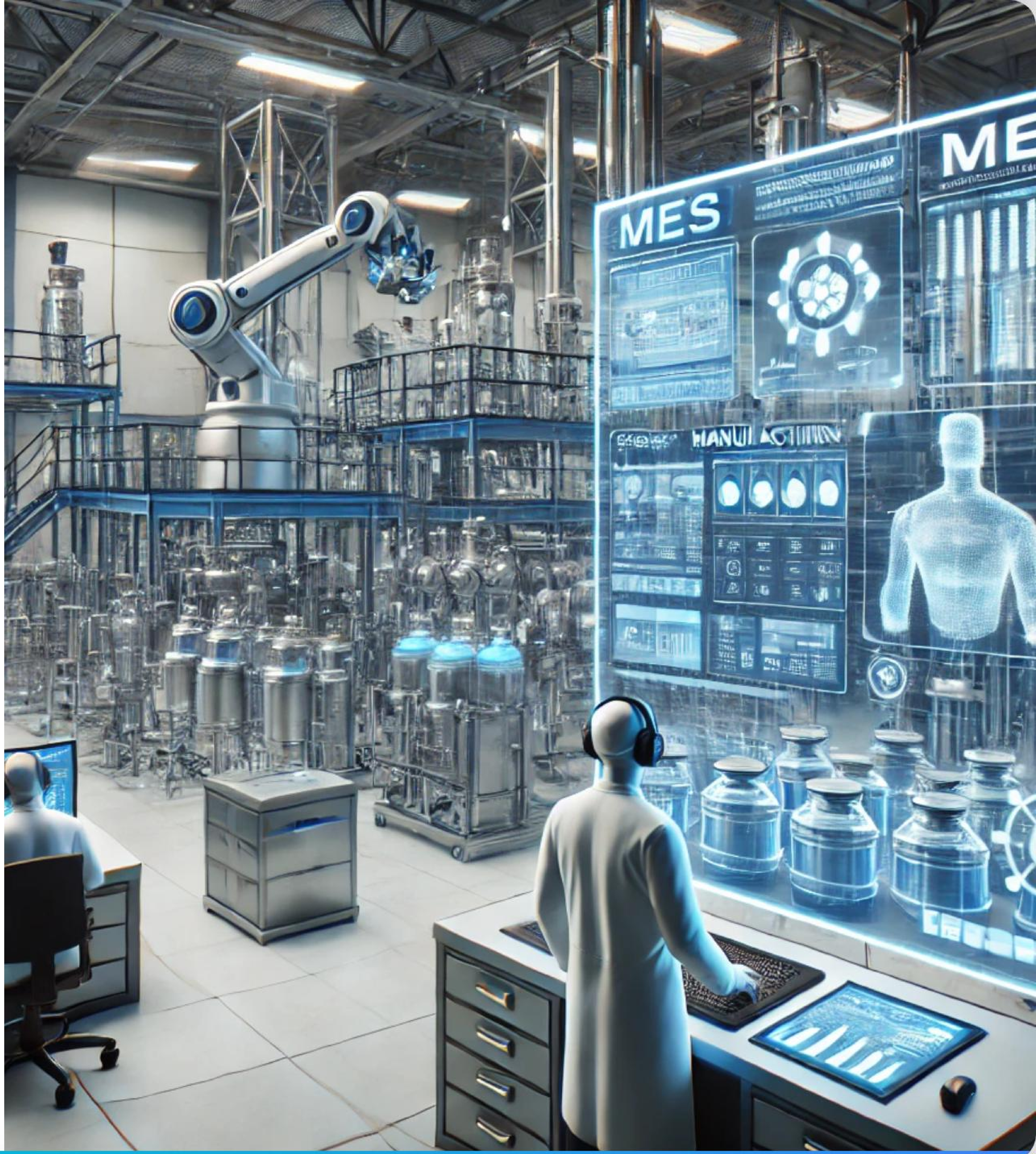
Category	Technologies & Tools	Project Purpose & Role
Language & Core	Python 3.10+, Jupyter Notebooks	End-to-end data processing, model engineering, & experimental execution.
Data Processing	Pandas, NumPy	Missing value audit, duplicate detection, matrix math, & feature transformations.
ML Algorithms	Scikit-Learn, XGBoost, CatBoost, LightGBM	Tree-based regression modeling, cross-validation, & hyperparameter optimization.
Optimization	RandomizedSearchCV, K-Fold CV	Systematic hyperparameter search & out-of-fold generalization validation.
Explainability & Viz	SHAP, Matplotlib, Seaborn	Model interpretability plots, feature importance ranking, & correlation heatmaps.

Expected Industrial Impact

Transforming Smart Manufacturing

Implementing machine learning surrogate models in continuous chemical plants delivers immediate tangible operational benefits:

- **Sub-Second Latency:** Replaces 30+ minute numerical solvers with instantaneous output.
- **Waste & Yield Optimization:** Reduces unwanted byproduct formation (Product C) by up to 15%.
- **Digital Twin Integration:** Plugs directly into plant SCADA systems for real-time closed-loop control.



Project Implementation Roadmap

Phase 1: Data & EDA

Cleaned dataset, audited distribution, confirmed zero missing/duplicate values, & built heatmaps.

Phase 2: Feature Eng.

Formulated domain features: Residence Time Index, Thermal Energy, & Delta T.

Phase 3: Optimization

Tuned Gradient Boosting & XGBoost via RandomizedSearchCV & K-Fold cross-validation.

Phase 4: SHAP & Deploy

SHAP interpretability, ensemble averaging, & generation of final unseen predictions.

Expected Impact & Value

10,000x

Faster Simulation Speed

- **Yield Improvement:** +3% to 5% higher conversion efficiency for Product B by maintaining optimal operational setpoints.
- **Energy & Waste Reduction:** Minimizes side-product C generation, reducing downstream purification costs and carbon footprint.
- **Smart Manufacturing:** Seamlessly enables real-time automated Closed-Loop Control in continuous chemical plants.

Scalability & Future Horizons

Our surrogate ML architecture is engineered for effortless scaling across complex multi-reactor networks and plant-wide chemical processes.

PINNs Integration

Embedding kinetic mass balances into neural loss functions for physical guarantees.

FastAPI & Docker

Containerized microservice API deployment for instant industrial IoT connectivity.

Online Active Learning

Continuous model retraining on streaming sensor telemetry to prevent model drift.

Thank You!

Industrial Chemical Reactor Yield Prediction Using Machine Learning

Team: Saubhagya se Prem | https://github.com/shiksha5245